

Installing and Configuring Postfix as a Send-only SMTP Server on Ubuntu 20.04

Postfix (http://www.postfix.org/) is a popular free and open-source software that acts as a mail transfer agent (MTA (https://www.techopedia.com/definition/1691/message-transfer-agent-mta)). The job of Postfix is to route and deliver emails. Besides the classic usage, Postfix can also be configured to send emails by local applications only.

This application of Postfix is a useful trick to deploy in various situations. For example, it can be used when sending email notifications regularly, working with a third-party email service provider with limited outbound traffic, etc. Compared to any full-fledge SMTP (https://www.techopedia.com/definition/1710/simple-mail-transfer-protocol-smtp) server, Postfix is a lighter alternative that at the same time retains the necessary functionalities.

In this tutorial, check out **how to install and configure Postfix as a send-only SMTP server on Ubuntu.**

We use cookies on our website to give you the most relevant experience by remembering your preferences and repeat visits. By clicking “Accept”, you consent to the use of ALL the cookies. However you may visit Cookie Settings to provide a controlled consent.

Cookie settings ACCEPT

Prerequisites

In order to follow this guide, you need to fulfill the following prerequisites:

- You need to have a properly configured Ubuntu server (https://www.cloudsigma.com/how-to-set-up-your-ubuntu-18-04-server/) with access to a sudo non-root user (https://www.cloudsigma.com/configuring-the-linux-sudoers-file/).
- You also need a fully registered domain name. Learn more about domain names from our Installing an SSL Certificate from a Commercial Certificate Authority (https://www.cloudsigma.com/installing-an-ssl-certificate-from-a-commercial-certificate-authority-a-comprehensive-guide/) post.
- Finally, you will need a DNS record with the domain name pointing towards the server's public IP address.

Installing Postfix

Once the preconditions are met, the server is ready to deploy Postfix. The simplest way of installing Postfix is to install the `mailutils` package. It's directly available from the official Ubuntu repos.

First, launch a terminal, and update the `APT` package database:

```
1 sudo apt update
```

```
cloudsigma@server:~$ sudo apt update
Hit:1 http://ch.archive.ubuntu.com/ubuntu focal InRelease
Hit:2 http://ch.archive.ubuntu.com/ubuntu focal-updates InRelease
Hit:3 http://ch.archive.ubuntu.com/ubuntu focal-backports InRelease
Hit:4 http://ch.archive.ubuntu.com/ubuntu focal-security InRelease
Hit:5 https://download.docker.com/linux/ubuntu focal InRelease
Hit:6 http://ppa.launchpad.net/deadsnakes/ppa/ubuntu focal InRelease
Hit:7 https://packages.cloud.google.com/apt kubernetes-xenial InRelease
Reading package lists... Done
Building dependency tree
Reading state information... Done
```

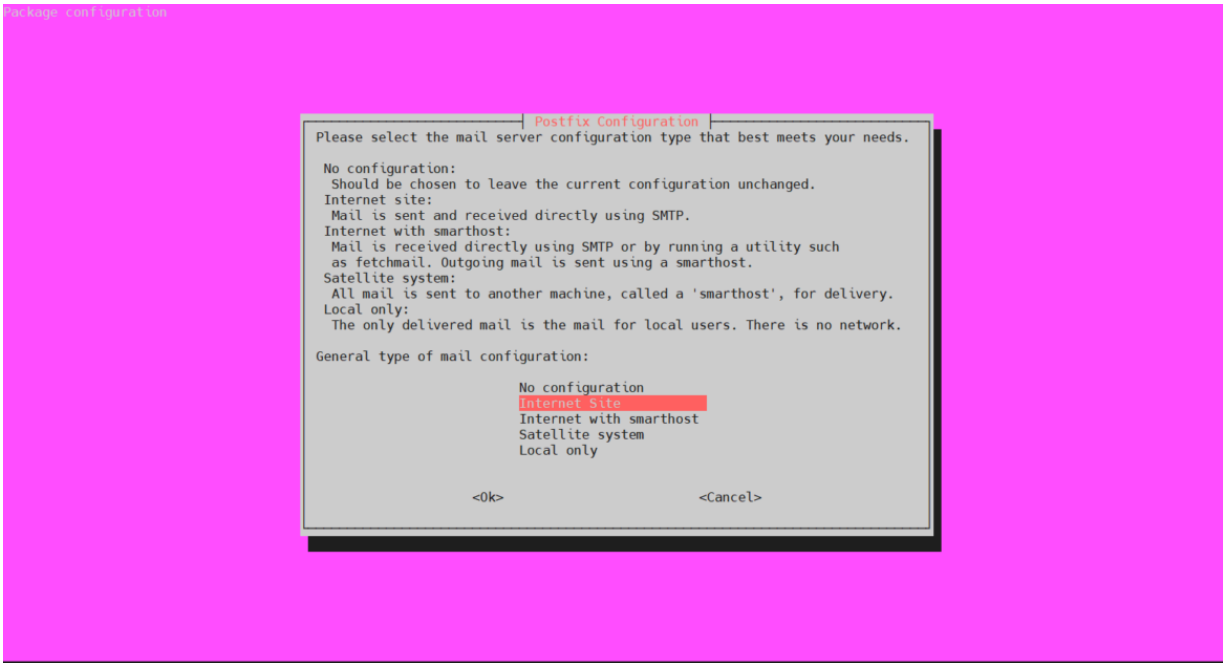
Then, you can install Postfix:

```
1 sudo apt install mailutils
```

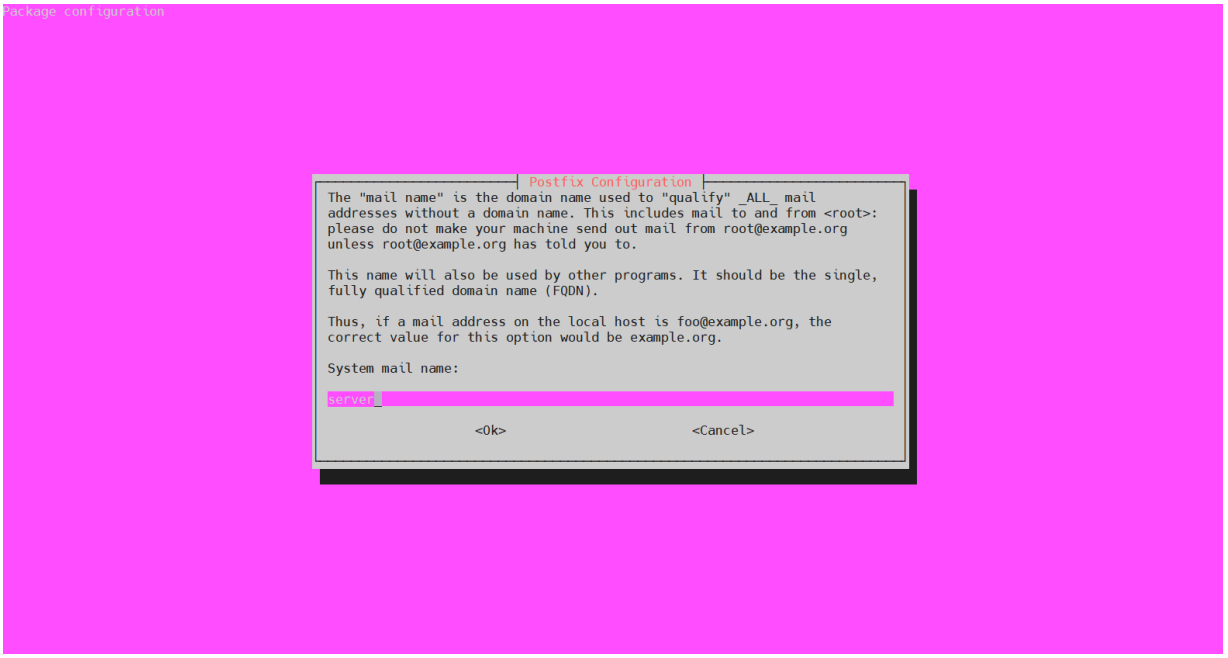
```
cloudsigma@server:~$ sudo apt install mailutils
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  guile-2.2-libs libgc1c2 libgsasl7 libidn11 libkyotocabinet16v5 libmailutils6 libmysqlclient21 libntlm0
Suggested packages:
  mailutils-mh mailutils-doc procmail postfix-mysql postfix-pgsql postfix-ldap postfix-pcre postfix-lmdb
  postfix-sqlite postfix-doc openssl-blacklist
The following NEW packages will be installed:
  guile-2.2-libs libgc1c2 libgsasl7 libidn11 libkyotocabinet16v5 libmailutils6 libmysqlclient21 libntlm0
  ssl-cert
0 upgraded, 13 newly installed, 0 to remove and 99 not upgraded.
Need to get 8,902 kB of archives.
After this operation, 64.3 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

We use cookies on our website to give you the most relevant experience by remembering your preferences and repeat visits. By clicking "Accept", you consent to the use of ALL the cookies. However you may visit Cookie Settings to provide a controlled consent.
Cookie settings ACCEPT

During the installation process, the following configuration window will pop up. The default option is `Internet Site`. To confirm the selection, press `TAB` to move the cursor to the next section. Then, hit `Enter`:



The next step will ask for the System mail name. It will be the name assigned to the server at the time of creation. As described in the script, if the mail address for the local host is "foo@example.org," then the System mail name would be `example.org`:



Note that the configuration script can also be triggered anytime later using the following command:

```
1 sudo dpkg-reconfigure postfix
```

We use cookies on our website to give you the most relevant experience by remembering your preferences and repeat visits. By clicking "Accept", you consent to the use of ALL the cookies. However you may visit Cookie Settings to provide a controlled consent.

Cookie settings ACCEPT

At this step, Postfix will be configured to send and receive emails only from `localhost`. This requires Postfix to listen on the `loopback` interface. It's the virtual network interface that the server uses for internal communication. Next, open the Postfix configuration file using your text editor of choice:

```
1 sudo nano /etc/postfix/main.cf
```

Change the value of `inet_interface` to `loopback-only`:

```
  GNU nano 4.8                          /etc/postfix/main.cf

# TLS parameters
smtpd_tls_cert_file=/etc/ssl/certs/ssl-cert-snakeoil.pem
smtpd_tls_key_file=/etc/ssl/private/ssl-cert-snakeoil.key
smtpd_tls_security_level=may

smtp_tls_CApath=/etc/ssl/certs
smtp_tls_security_level=may
smtp_tls_session_cache_database = btree:${data_directory}/smtp_scache


smtpd_relay_restrictions = permit_mynetworks permit_sasl_authenticated defer_unauth_destination
myhostname = server
alias_maps = hash:/etc/aliases
alias_database = hash:/etc/aliases
mydestination = $myhostname, server, localhost.localdomain, , localhost
relayhost =
mynetworks = 127.0.0.0/8 [::ffff:127.0.0.0]/104 [::1]/128
mailbox_size_limit = 0
recipient_delimiter = +
inet_interfaces = loopback-only
inet_protocols = all



^G Get Help     ^O Write Out    ^W Where Is     ^K Cut Text     ^J Justify      ^C Cur Pos
^X Exit         ^R Read File    ^\ Replace      ^U Paste Text   ^T To Spell     ^_ Go To Line
```

Another directive that you have to change is `mydestination`. It defines the list of domains that are delivered via the `local_transport` mail delivery transport. The default value will be something like this:

```
  GNU nano 4.8                          /etc/postfix/main.cf


# TLS parameters
smtpd_tls_cert_file=/etc/ssl/certs/ssl-cert-snakeoil.pem
smtpd_tls_key_file=/etc/ssl/private/ssl-cert-snakeoil.key
smtpd_tls_security_level=may

smtp_tls_CApath=/etc/ssl/certs
smtp_tls_security_level=may
smtp_tls_session_cache_database = btree:${data_directory}/smtp_scache


smtpd_relay_restrictions = permit_mynetworks permit_sasl_authenticated defer_unauth_destination
myhostname = server
alias_maps = hash:/etc/aliases
alias_database = hash:/etc/aliases
mydestination = $myhostname, server, localhost.localdomain, , localhost
relayhost =
mynetworks = 127.0.0.0/8 [::ffff:127.0.0.0]/104 [::1]/128
mailbox_size_limit = 0
recipient_delimiter = +
inet_interfaces = loopback-only
inet_protocols = all

^G Get Help     ^O Write Out    ^W Where Is     ^K Cut Text     ^J Justify      ^C Cur Pos
^X Exit         ^R Read File    ^\ Replace      ^U Paste Text   ^T To Spell     ^_ Go To Line
```

We use cookies on our website to give you the most relevant experience by remembering your preferences and repeat visits. By clicking "Accept", you consent to the use of ALL the cookies. However you may visit Cookie Settings to provide a controlled consent.

Cookie settings ACCEPT

Change its value to the following:

```
mydestination = localhost.$mydomain, localhost, $myhostname
```

```
  GNU nano 4.8                          /etc/postfix/main.cf

# See http://www.postfix.org/COMPATIBILITY_README.html -- default to 2 on
# fresh installs.
compatibility_level = 2



# TLS parameters
smtpd_tls_cert_file=/etc/ssl/certs/ssl-cert-snakeoil.pem
smtpd_tls_key_file=/etc/ssl/private/ssl-cert-snakeoil.key
smtpd_tls_security_level=may

smtp_tls_CApath=/etc/ssl/certs
smtp_tls_security_level=may
smtp_tls_session_cache_database = btree:${data_directory}/smtp_scache


smtpd_relay_restrictions = permit_mynetworks permit_sasl_authenticated defer_unauth_destination
myhostname = server
alias_maps = hash:/etc/aliases
alias_database = hash:/etc/aliases
mydestination = localhost.$mydomain, localhost, $myhostname
relayhost =
mynetworks = 127.0.0.0/8 [::ffff:127.0.0.0]/104 [::1]/128

^G Get Help     ^O Write Out    ^W Where Is     ^K Cut Text     ^J Justify      ^C Cur Pos
^X Exit         ^R Read File    ^\ Replace      ^U Paste Text   ^T To Spell     ^_ Go To Line
```

In case your domain is actually a subdomain and you want the emails to appear as if sent from the main domain, add the following directive at the end of the Postfix configuration. It will remove the subdomain from the email address:

```
masquerade_domains = <main_domain>
```

```
  GNU nano 4.8                          /etc/postfix/main.cf
# TLS parameters
smtpd_tls_cert_file=/etc/ssl/certs/ssl-cert-snakeoil.pem
smtpd_tls_key_file=/etc/ssl/private/ssl-cert-snakeoil.key
smtpd_tls_security_level=may

smtp_tls_CApath=/etc/ssl/certs
smtp_tls_security_level=may
smtp_tls_session_cache_database = btree:${data_directory}/smtp_scache


smtpd_relay_restrictions = permit_mynetworks permit_sasl_authenticated defer_unauth_destination
myhostname = server
alias_maps = hash:/etc/aliases
alias_database = hash:/etc/aliases
mydestination = localhost.$mydomain, localhost, $myhostname
relayhost =
mynetworks = 127.0.0.0/8 [::ffff:127.0.0.0]/104 [::1]/128
mailbox_size_limit = 0
recipient_delimiter = +
inet_interfaces = loopback-only
inet_protocols = all

masquerade_domains = my_domain


^G Get Help     ^O Write Out    ^W Where Is     ^K Cut Text     ^J Justify      ^C Cur Pos
^X Exit         ^R Read File    ^\ Replace      ^U Paste Text   ^T To Spell     ^_ Go To Line
```

To take the changes into effect, restart Postfix:

```
sudo systemctl restart postfix
```

We use cookies on our website to give you the most relevant experience by remembering your preferences and repeat visits. By clicking “Accept”, you consent to the use of ALL the cookies. However you may visit Cookie Settings to provide a controlled consent.

Cookie settings ACCEPT

```
cloudsigma@server:~$ sudo systemctl restart postfix
cloudsigma@server:~$
```

Testing the SMTP Server

Assuming the configuration process went successfully, it's time to test it. In order to do that, check whether Postfix can send an email to an external email account using the `mail` command. It's a part of the `mailutils` package:

```
1 echo "Test email body" | mail -s "Subject line" <email_address>
```

The message should arrive at the email address specified. At this point, all the emails sent are unencrypted. Service providers will generally tag such emails as spam, so ensure to check the spam section for the email's arrival.

If the `mail` command reports an error or the email didn't arrive after a prolonged period of time, re-check the Postfix configuration and ensure that the server name and hostname are set properly. With this configuration, the emails will appear to be sent from the following address:

```
1 <username>@<domain>
```

Here, the username will be the username of the server user that ran the `mail` command.

Forwarding System Mail

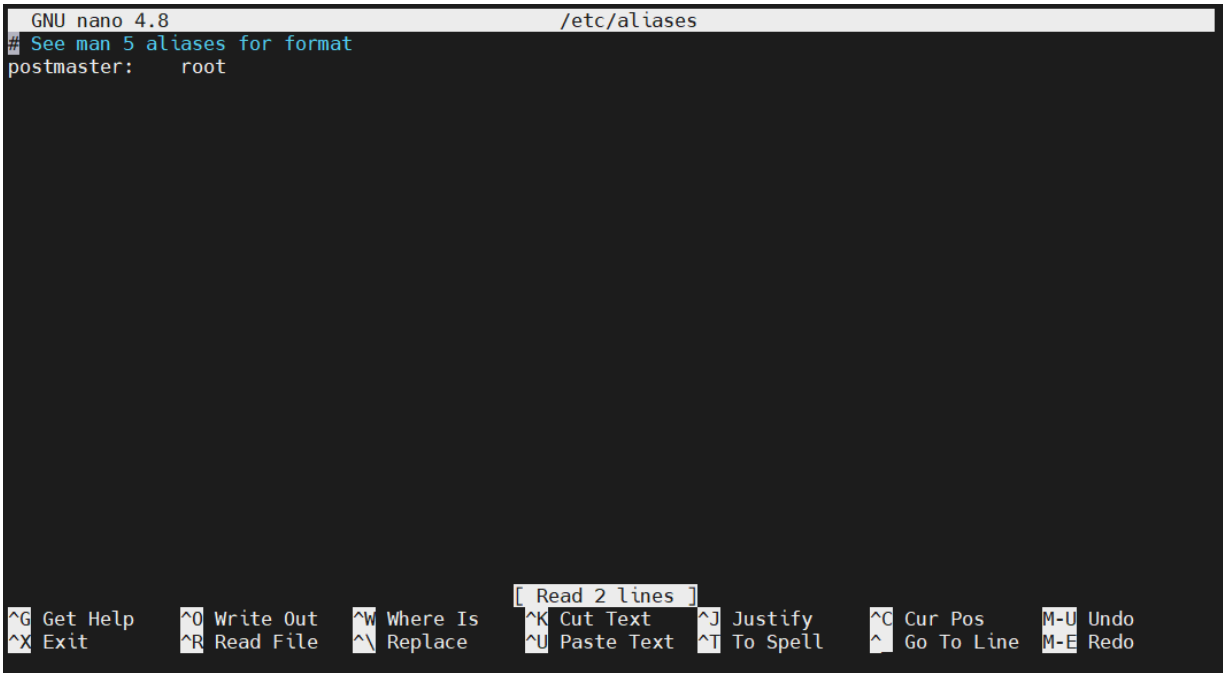
We've successfully verified that the email server is properly configured and functional. Now it has to be configured to forward email for `root`. All the system-generated messages sent to the server will get forwarded to an external email address. For email recipients, the `/etc/aliases` file contains the list of alternate names. Open it using your text editor of choice:

```
1 sudo nano /etc/aliases
```

By default, it will look like this:

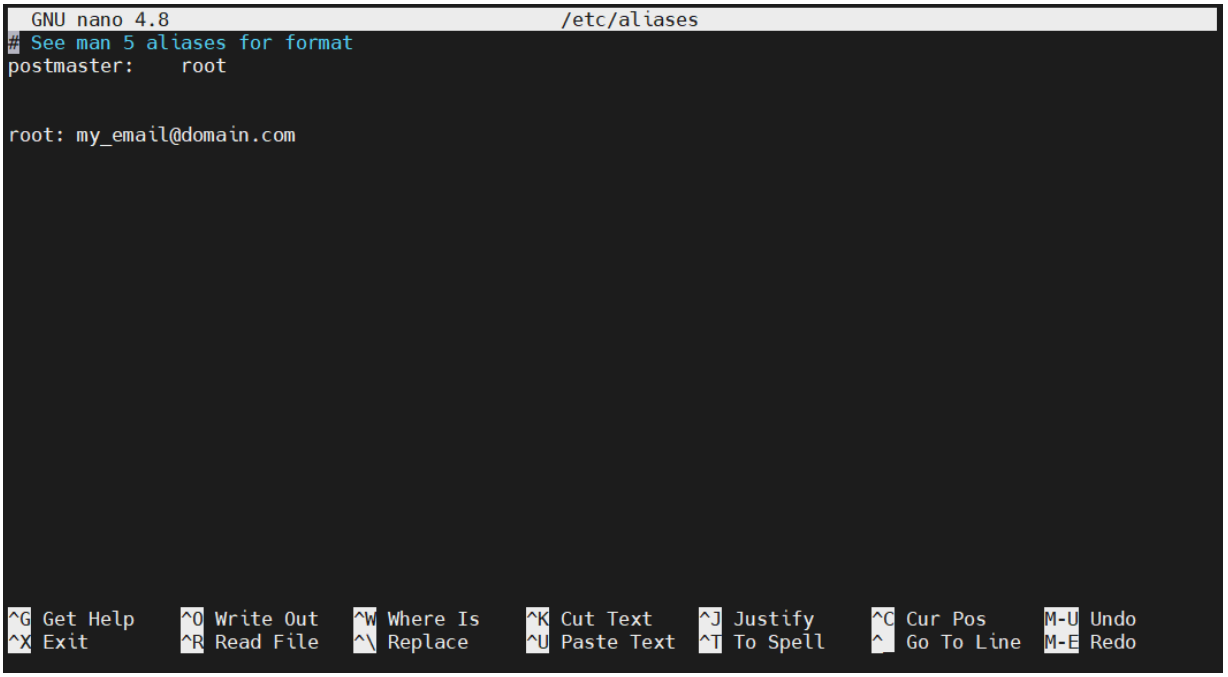
We use cookies on our website to give you the most relevant experience by remembering your preferences and repeat visits. By clicking "Accept", you consent to the use of ALL the cookies. However you may visit Cookie Settings to provide a controlled consent.

Cookie settings ACCEPT



At the end of the file, add the following line:

```
root: <email_address>
```



It tells that the emails sent to `root` will be forwarded to an email address. To take the change into effect, run the following command. It will rebuild the alias database that the `mail` command uses:

```
sudo newaliases
```

```
cloudsigma@server:~$ sudo newaliases
cloudsigma@server:~$
```

Next, you need to test if the change was successful. Send a test email to `root`:

```
echo "Test root email body" | mail -s "Subject line" root
```

We use cookies on our website to give you the most relevant experience by remembering your preferences and repeat visits. By clicking “Accept”, you consent to the use of ALL the cookies. However you may visit Cookie Settings to provide a controlled consent.

Cookie settings ACCEPT

The email should arrive at the specified email address. Don't forget to check the spam folder as there's no encryption configured yet.

Enabling SMTP Encryption

Sending emails in unencrypted format is a risky move. To ensure safety, it's required to have SMTP encryption enabled. To achieve this, we'll be requesting a free TLS certificate from Let's Encrypt (https://letsencrypt.org/) for your domain.

For Ubuntu, it's the Certbot (https://certbot.eff.org/) tool that does the job. Thankfully, it's directly available from the official Ubuntu repo. Install Certbot with the following command:

```
1 sudo apt install certbot
```

```
cloudsigma@server:~$ sudo apt install certbot
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  python3-acme python3-certbot python3-configargparse python3-future python3-icu python3-josepy pyt
  python3-requests-toolbelt python3-rfc3339 python3-tz python3-zope.component python3-zope.event py
Suggested packages:
  python3-certbot-apache python3-certbot-nginx python-certbot-doc python-acme-doc python-future-doc
The following NEW packages will be installed:
  certbot python3-acme python3-certbot python3-configargparse python3-future python3-icu python3-jo
  python3-requests-toolbelt python3-rfc3339 python3-tz python3-zope.component python3-zope.event py
0 upgraded, 16 newly installed, 0 to remove and 101 not upgraded.
Need to get 1,152 kB of archives.
After this operation, 6,066 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://ch.archive.ubuntu.com/ubuntu focal/universe amd64 python3-josepy all 1.2.0-2 [28.1 kB]
Get:2 http://ch.archive.ubuntu.com/ubuntu focal/main amd64 python3-pbr all 5.4.5-0ubuntu1 [64.0 kB]
Get:3 http://ch.archive.ubuntu.com/ubuntu focal/universe amd64 python3-mock all 3.0.5-1build1 [25.6
Get:4 http://ch.archive.ubuntu.com/ubuntu focal/universe amd64 python3-requests-toolbelt all 0.8.0-
Get:5 http://ch.archive.ubuntu.com/ubuntu focal/main amd64 python3-tz all 2019.3-1 [24.4 kB]
Get:6 http://ch.archive.ubuntu.com/ubuntu focal/main amd64 python3-rfc3339 all 1.1-2 [6,808 B]
Get:7 http://ch.archive.ubuntu.com/ubuntu focal/universe amd64 python3-acme all 1.1.0-1 [29.6 kB]
Get:8 http://ch.archive.ubuntu.com/ubuntu focal/universe amd64 python3-configargparse all 0.13.0-2
Get:9 http://ch.archive.ubuntu.com/ubuntu focal/main amd64 python3-future all 0.18.2-2 [336 kB]
Get:10 http://ch.archive.ubuntu.com/ubuntu focal/universe amd64 python3-parsedatetime all 2.4-5 [32
Get:11 http://ch.archive.ubuntu.com/ubuntu focal/universe amd64 python3-zope.hookable amd64 5.0.0-1
```

Assuming the server has UFW configured (https://www.cloudsigma.com/the-basics-of-ufw-learning-the-essential-firewall-commands/), it's necessary to have port `80` open so that domain verification can take place. Run the following command to enable it:

```
1 sudo ufw allow 80
```

```
cloudsigma@server:~$ sudo ufw allow 80
Rules updated
Rules updated (v6)
```

Now, run Certbot to generate a certificate. It commands Certbot to issue certificates with a 4096-bit RSA key. The verification will take place via port `80` (HTTP):

```
1 sudo certbot certonly --standalone --rsa-key-size 4096 --agree-tos --preferred-ch
```

The output will be something like this:

We use cookies on our website to give you the most relevant experience by remembering your preferences and repeat visits. By clicking "Accept", you consent to the use of ALL the cookies. However you may visit Cookie Settings to provide a controlled consent.

Cookie settings ACCEPT

```
Saving debug log to /var/log/letsencrypt/letsencrypt.log

Plugins selected: Authenticator standalone, Installer None

Obtaining a new certificate

Performing the following challenges:

http-01 challenge for '<domain>'

Waiting for verification...

Cleaning up challenges

IMPORTANT NOTES:

- Congratulations! Your certificate and chain have been saved at:

/etc/letsencrypt/live/<domain>/fullchain.pem

Your key file has been saved at:

/etc/letsencrypt/live/<domain>/privkey.pem

Your cert will expire on 2020-07-11. To obtain a new or tweaked

version of this certificate in the future, simply run certbot

again. To non-interactively renew *all* of your certificates, run

"certbot renew"

- If you like Certbot, please consider supporting our work by:

Donating to ISRG / Let's Encrypt: https://letsencrypt.org/donate

Donating to EFF: https://eff.org/donate-le
```

The key will be stored in the following directory:

```
/etc/letsencrypt/live/<domain>
```

Now that your certificate is generated, open the Postfix configuration file in a text editor:

```
sudo nano /etc/postfix/main.cf
```

Find the section `TLS parameters`:

```
  GNU nano 4.8                          /etc/postfix/main.cf


# TLS parameters
smtpd_tls_cert_file=/etc/ssl/certs/ssl-cert-snakeoil.pem
smtpd_tls_key_file=/etc/ssl/private/ssl-cert-snakeoil.key
smtpd_tls_security_level=may

smtp_tls_CApath=/etc/ssl/certs
smtp_tls_security_level=may
smtp_tls_session_cache_database = btree:${data_directory}/smtp_scache
```

After that, change the value of the directives `smtpd_tls_cert_file` and `smtpd_tls_key_file`:

We use cookies on our website to give you the most relevant experience by remembering your preferences and repeat visits. By clicking “Accept”, you consent to the use of ALL the cookies. However you may visit Cookie Settings to provide a controlled consent.

Cookie settings ACCEPT

```
  GNU nano 4.8                          /etc/postfix/main.cf


# TLS parameters
smtpd_tls_cert_file=/etc/letsencrypt/live/<domain>/fullchain.pem
smtpd_tls_key_file=/etc/letsencrypt/live/<domain>/privkey.pem
smtpd_tls_security_level=may

smtp_tls_CApath=/etc/ssl/certs
smtp_tls_security_level=may
smtp_tls_session_cache_database = btree:${data_directory}/smtp_scache
```

Next, save and close the file. To take the changes into effect, restart Postfix:

```
1 sudo systemctl restart postfix
```

Finally, send a dummy email again to a target email address:

```
1 echo "This is email body" | mail -s "Email subject line" <email>
```

The email should now appear normally like other emails. If you check the technical info of the email, it will appear encrypted.

Final Thoughts

Congratulations! You've successfully configured a send-only email server, powered by Postfix. The email transactions are also secure using an appropriate encryption key.

For more ways to optimize email transfer check out this tutorial The Best Ways to Use Google's SMTP Server (https://www.cloudsigma.com/the-best-ways-to-use-googles-smtp-server/).

Happy computing!

About | Latest



About Pranay Kapgate

Software Engineer | Open Source Software aficionado

CLOUD (HTTPS://WWW.CLOUDSIGMA.COM/TAG/CLOUD/)

CLOUD TUTORIALS (HTTPS://WWW.CLOUDSIGMA.COM/TAG/CLOUD-TUTORIALS/)

CONFIGURE POSTFIX (HTTPS://WWW.CLOUDSIGMA.COM/TAG/CONFIGURE-POSTFIX/)

We use cookies on our website to give you the most relevant experience by remembering your preferences and repeat visits. By clicking "Accept", you consent to the use of ALL the cookies. However you may visit Cookie Settings to provide a controlled consent.

Cookie settings ACCEPT

EMAIL TRANSFER (HTTPS://WWW.CLOUDSIGMA.COM/TAG/EMAIL-TRANSFER/)

INSTALL POSTFIX (HTTPS://WWW.CLOUDSIGMA.COM/TAG/INSTALL-POSTFIX/)

MAIL TRANSFER AGENT (HTTPS://WWW.CLOUDSIGMA.COM/TAG/MAIL-TRANSFER-AGENT/)

MTA (HTTPS://WWW.CLOUDSIGMA.COM/TAG/MTA/)

OPEN-SOURCE SOFTWARE (HTTPS://WWW.CLOUDSIGMA.COM/TAG/OPEN-SOURCE-SOFTWARE/)

POSTFIX (HTTPS://WWW.CLOUDSIGMA.COM/TAG/POSTFIX/)

SEND-ONLY SMTP SERVER (HTTPS://WWW.CLOUDSIGMA.COM/TAG/SEND-ONLY-SMTP-SERVER/)

SMTP (HTTPS://WWW.CLOUDSIGMA.COM/TAG/SMTP/)

SMTP ENCRYPTION (HTTPS://WWW.CLOUDSIGMA.COM/TAG/SMTP-ENCRYPTION/)

SMTP SERVER (HTTPS://WWW.CLOUDSIGMA.COM/TAG/SMTP-SERVER/)

UBUNTU (HTTPS://WWW.CLOUDSIGMA.COM/TAG/UBUNTU/)

Search

Search

Recent Posts

Overview of DNS Terminologies, Components, and Concepts (https://www.cloudsigma.com/overview-of-dns-terminologies-components-and-concepts/)

Defining Functions in Python 3: A Tutorial (https://www.cloudsigma.com/defining-functions-in-python-3-a-tutorial/)

Troubleshooting Common HTTP Error Codes (https://www.cloudsigma.com/troubleshooting-common-http-error-codes/)

Receive a Weekly Digest

First Name

Your first name

Email address:

Your email address

We use cookies on our website to give you the most relevant experience by remembering your preferences and repeat visits. By clicking “Accept”, you consent to the use of ALL the cookies. However you may visit Cookie Settings to provide a controlled consent.

Cookie settings ACCEPT

SUBSCRIBE TO OUR LATEST BLOGS

(https://www.facebook.com/CloudSigma)
(https://twitter.com/CloudSigma)
(https://www.linkedin.com/company/cloudsigma-ag)
(https://www.youtube.com/cloudsigma)
(https://www.instagram.com/cloudsigmahq/)
(https://www.cloudsigma.com/feed/)

(https://marketplace.intel.com/s/partner/a5S3b0000002kE0EAI/cloudsigma?language=en_US&wapkw=cloudsigma)

(https://www.cloudsigma.com/cloudsigma-with-hpe-delivers-innovative-cloud-services-around-the-globe/)

(https://www.cloudsigma.com/iso-27001-information-security-certified-cloud/)

We use cookies on our website to give you the most relevant experience by remembering your preferences and repeat visits. By clicking “Accept”, you consent to the use of ALL the cookies. However you may visit Cookie Settings to provide a controlled consent.

Cookie settings ACCEPT

(https://www.cloudsigma.com/iso-27017-cloud-security-certified-cloud/)

(https://www.cloudsigma.com/iso-27018-privacy-protection-certified-cloud/)

(https://www.cloudsigma.com/geant-preferred-cloud-partner//)

(https://www.cloudsigma.com/eu-gdpr-compliant-cloud/)

(https://www.cloudsigma.com/pci-dss-compliant-cloud/) (https://www.cloudsigma.com

/star-level-1-registered-public-cloud/) (https://www.cloudsigma.com/eba-

recommendations-compliant-cloud/)

First Name

Your first name

Email address:

We use cookies on our website to give you the most relevant experience by remembering your preferences and repeat visits. By clicking “Accept”, you consent to the use of ALL the cookies. However you may visit Cookie Settings to provide a controlled consent.

Cookie settings ACCEPT

Your email address

SUBSCRIBE TO OUR LATEST BLOGS

© 2022 CloudSigma AG

We use cookies on our website to give you the most relevant experience by remembering your preferences and repeat visits. By clicking “Accept”, you consent to the use of ALL the cookies. However you may visit Cookie Settings to provide a controlled consent.

Cookie settings ACCEPT